

CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Business Analytics Project Report

Prepared as part of the Business Analytics Internship Project

Tools Used: MySQL, Power BI and Excel

Dataset: Customer Shopping Behavior Dataset

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Internship Organization: Wahy Lab Solutions

Project: Customer Shopping Behavior Analysis

Project Type: Business Analytics Internship Project

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1. Introduction

Retail businesses generate large amounts of customer and purchase information. Analysing this information helps management understand what customers buy, how frequently they purchase, which categories generate sales, and which marketing factors are associated with purchasing behaviour.

This project analyses a customer shopping behaviour dataset using SQL for data preparation and analysis and Power BI for interactive visualization. The objective is to convert raw customer-level shopping records into useful business insights that can support customer engagement, product planning, marketing decisions and loyalty initiatives.

2. Business Problem

The business problem focuses on understanding customer shopping behaviour in order to improve sales, customer satisfaction and long-term loyalty. The problem statement highlights changes in purchasing patterns across demographics and product categories and asks the analysis to consider factors such as discounts, reviews, seasons, payment preferences and repeat purchases.

The overarching business question is: How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?

The project deliverables specified in the business problem are SQL-based data analysis and modelling, an interactive Power BI dashboard, a report and presentation, and a structured GitHub repository containing the project files.

3. Objectives

- Clean and organize the customer shopping dataset for analysis.
- Use SQL to calculate sales, purchase, customer and product-level metrics.
- Analyse customer behaviour across gender, age, category, season, location and subscription status.
- Create customer segments based on previous purchase behaviour.
- Evaluate discounts, promotional codes, payment methods and shipping preferences as purchase drivers.
- Build an interactive Power BI dashboard for management-level decision making.
- Develop practical recommendations for marketing, product planning and customer retention.

4. Dataset Description

The dataset contains 3,900 customer shopping records and 18 original fields. Each record represents a customer shopping transaction/observation with demographic, product, purchase and behavioural attributes.

Field	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Customer gender
Item Purchased	Product purchased
Category	Product category
Purchase Amount (USD)	Transaction purchase amount
Location	Customer location
Size	Product size
Color	Product colour
Season	Season associated with the purchase
Review Rating	Customer review rating
Subscription Status	Subscription status

Shipping Type	Shipping method
Discount Applied	Whether a discount was applied
Promo Code Used	Whether a promotional code was used
Previous Purchases	Number of previous purchases
Payment Method	Payment method used
Frequency of Purchases	Purchase frequency

5. Data Cleaning & Preparation

The raw dataset was loaded into MySQL and structured as the Customer_Shopping_Data table. The dataset contains 3,900 records. Review Rating contains 37 missing observations; the missing values were retained rather than assigning an unsupported rating.

The SQL workflow was organized into separate files for data cleaning, analysis and customer segmentation. This structure makes the project easier to review and reproduce.

- Created the Customer_Shopping_Data table with appropriate data types.
- Loaded all 3,900 records into MySQL.
- Checked the dataset structure and available fields.
- Used aggregate SQL functions such as SUM, AVG and COUNT for analysis.
- Created a Customer Segment field using Previous Purchases.
- Grouped customers into New, Regular and Loyal customer segments.

6. SQL Analysis

SQL was used to answer the main business questions and generate measures later reflected in the Power BI dashboard. The analysis covered overall sales, category performance, seasonal patterns, customer segments, gender, payment methods, promotions, discounts, subscription status, purchase frequency and review ratings.

Metric	Result
Total Sales	\$233,081
Total Purchases	3,900
Average Purchase	\$59.76
Average Previous Purchases	25.35
Average Review Rating	3.75 / 5 (based on available ratings)

6.1 Category Performance

Category	Total Sales
Clothing	\$104,264
Accessories	\$74,200
Footwear	\$36,093
Outerwear	\$18,524

Clothing generated the highest sales among the four categories, followed by Accessories.

6.2 Seasonal Performance

Season	Total Sales
Fall	\$60,018
Spring	\$58,679
Winter	\$58,607
Summer	\$55,777

Fall recorded the highest sales, while Summer recorded the lowest sales among the four seasons.

6.3 Gender Analysis

Gender	Records	Total Sales	Average Purchase
Male	2,652	\$157,890	\$59.54
Female	1,248	\$75,191	\$60.25

Male customers account for a larger share of the records and total sales in this dataset. Average purchase values are relatively close between the two groups.

7. Customer Segmentation

Customer segmentation was created using the Previous Purchases field. The project classifies customers with 1–10 previous purchases as New Customers, 11–30 as Regular Customers, and 31–50 as Loyal Customers.

Customer Segment	Customers	Total Sales	Average Purchase
New Customer	784	\$47,564	\$60.67
Regular Customer	1,567	\$92,533	\$59.05
Loyal Customer	1,549	\$92,984	\$60.03

Regular and Loyal Customers contribute the largest sales amounts. This indicates that retention and repeat-purchase strategies are important opportunities for the business.

8. Power BI Dashboard

A four-page interactive Power BI dashboard was developed to present the analysis in a management-friendly format. Slicers allow users to explore the data by relevant customer, product, seasonal and marketing attributes. The dashboard pages are presented below in the same order as the project.



Figure 1: Executive Overview Dashboard

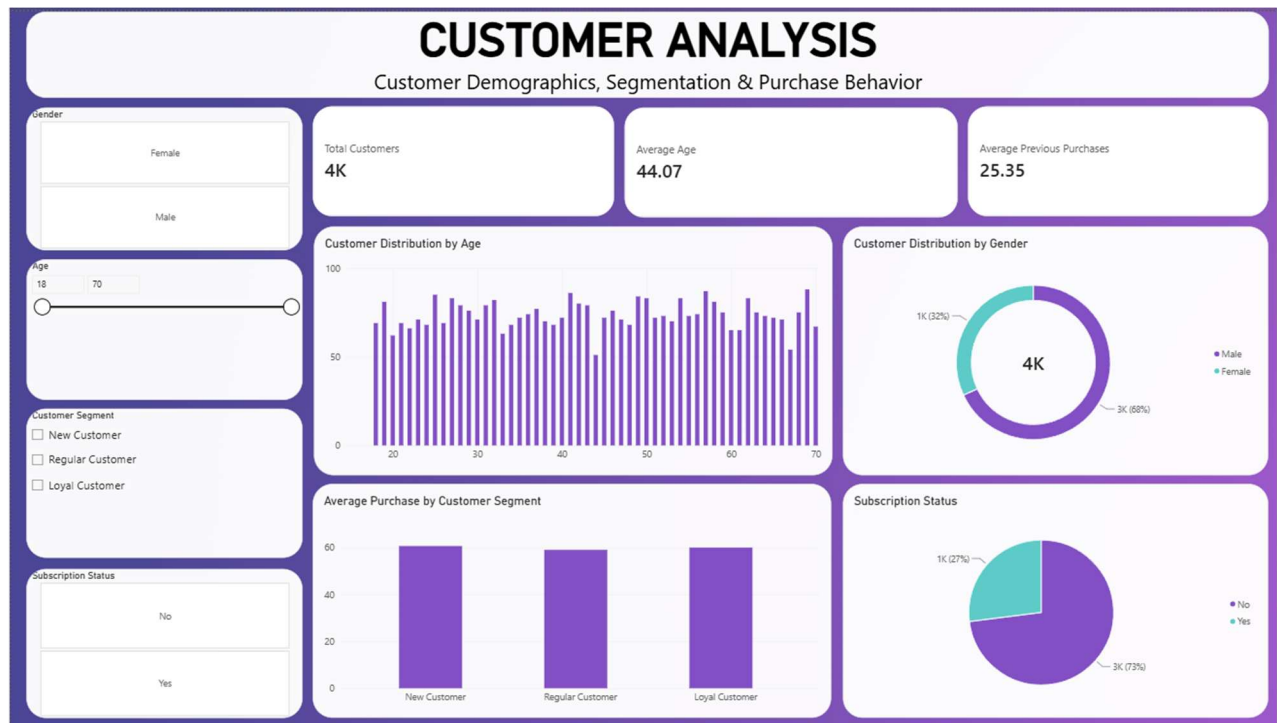


Figure 2: Customer Analysis Dashboard

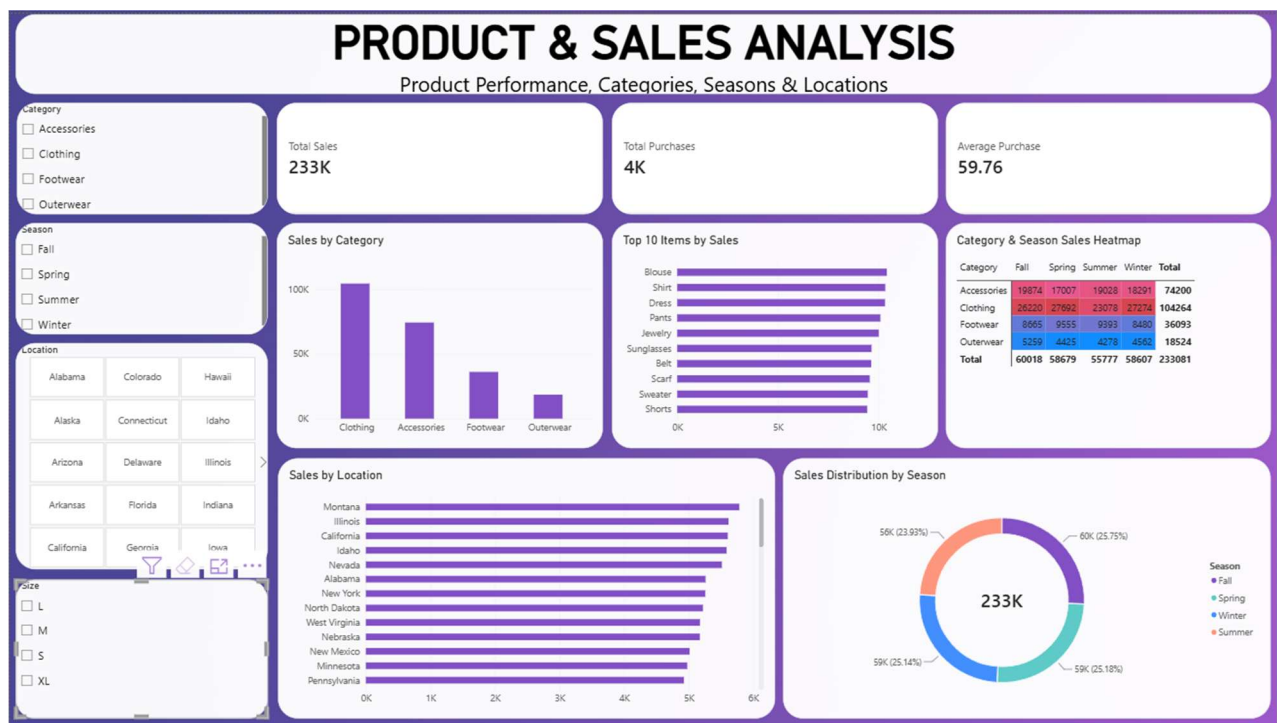


Figure 3: Product & Sales Analysis Dashboard

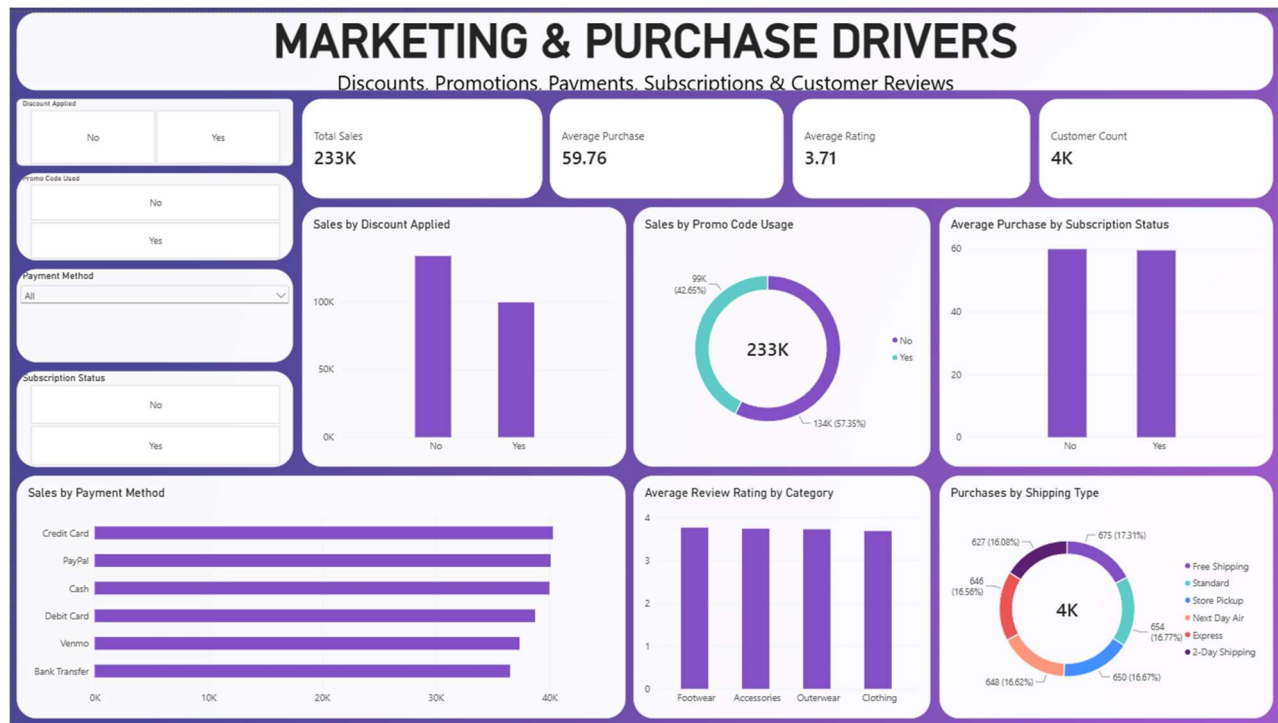


Figure 4: Marketing & Purchase Drivers Dashboard

9. Key Findings

- The dataset contains 3,900 purchase records with total sales of \$233,081.
- The average purchase amount is approximately \$59.76.
- Clothing is the strongest sales category with \$104,264 in sales.
- Accessories is the second-highest sales category with \$74,200.
- Fall has the highest seasonal sales at \$60,018.
- Regular and Loyal Customers together contribute the majority of sales in the customer segmentation analysis.
- Male customers account for more records and total sales, while average purchase values are similar across genders.
- Transactions without discounts generated slightly higher average purchase values than discounted transactions in this dataset.
- Payment methods are relatively distributed across Credit Card, PayPal, Cash, Debit Card, Venmo and Bank Transfer.
- The average available review rating is approximately 3.75 out of 5.

10. Business Recommendations

- Strengthen Clothing category planning: Maintain strong inventory availability and targeted campaigns for the highest-performing category.
- Focus on customer retention: Develop loyalty benefits, repeat-purchase offers and personalized communication for Regular and Loyal Customers.
- Use targeted promotions: Since discounted transactions show a slightly lower average purchase value, avoid relying only on blanket discounts and test targeted offers.
- Plan seasonal campaigns: Use Fall, Spring and Winter demand patterns to plan inventory and marketing activities before peak periods.

- Promote high-performing products: Use top-selling items as anchor products in campaigns and cross-selling strategies.
- Monitor customer satisfaction: Track review ratings by category and investigate categories or products with comparatively lower ratings.
- Support multiple payment preferences: Continue offering the payment methods used by customers while monitoring changes in payment behaviour.
- Use dashboard-based decision making: Review the Power BI dashboard regularly to identify changes in sales, customer segments and marketing performance.

11. Limitations

- The available dataset does not contain a transaction date, so time-series growth and month-by-month trend analysis cannot be performed from the supplied fields.
- The business problem mentions online versus offline channels, but a separate sales-channel field is not present in the supplied dataset; therefore, channel comparison was not performed.
- Customer Segment is a project-defined classification based on Previous Purchases and should be treated as an analytical segmentation rule rather than a formal CRM classification.
- The analysis identifies associations in the available data; it does not establish causal relationships between discounts, promotions and purchase behaviour.

12. Conclusion

This project demonstrates how SQL and Power BI can be combined to transform customer shopping data into actionable business insights. The analysis identifies the strongest product categories, seasonal patterns, customer segments, payment preferences and marketing-related patterns.

The results show that Clothing is the leading category and that Regular and Loyal Customers represent important revenue segments. The findings support a strategy focused on category strength, customer retention, targeted promotions, seasonal planning and continuous monitoring of customer satisfaction.

The final Power BI dashboard provides an interactive way for stakeholders to explore these patterns and use the results for data-driven marketing and product decisions.

13. Project Deliverables

- SQL – Data Cleaning queries
- SQL – Business Analysis queries
- SQL – Customer Segmentation queries
- Power BI – Customer Shopping Behavior Dashboard
- Project Report – Customer Shopping Behavior Analysis
- Presentation – Project findings and recommendations
- GitHub Repository – Structured project files

Note: The report is based on the supplied business problem statement and customer shopping behaviour dataset. All monetary values are in USD.

Power BI Dashboard Screenshots

The following screenshots show the four interactive dashboard pages developed for this project. They correspond to the dashboard sections described in Section 8.

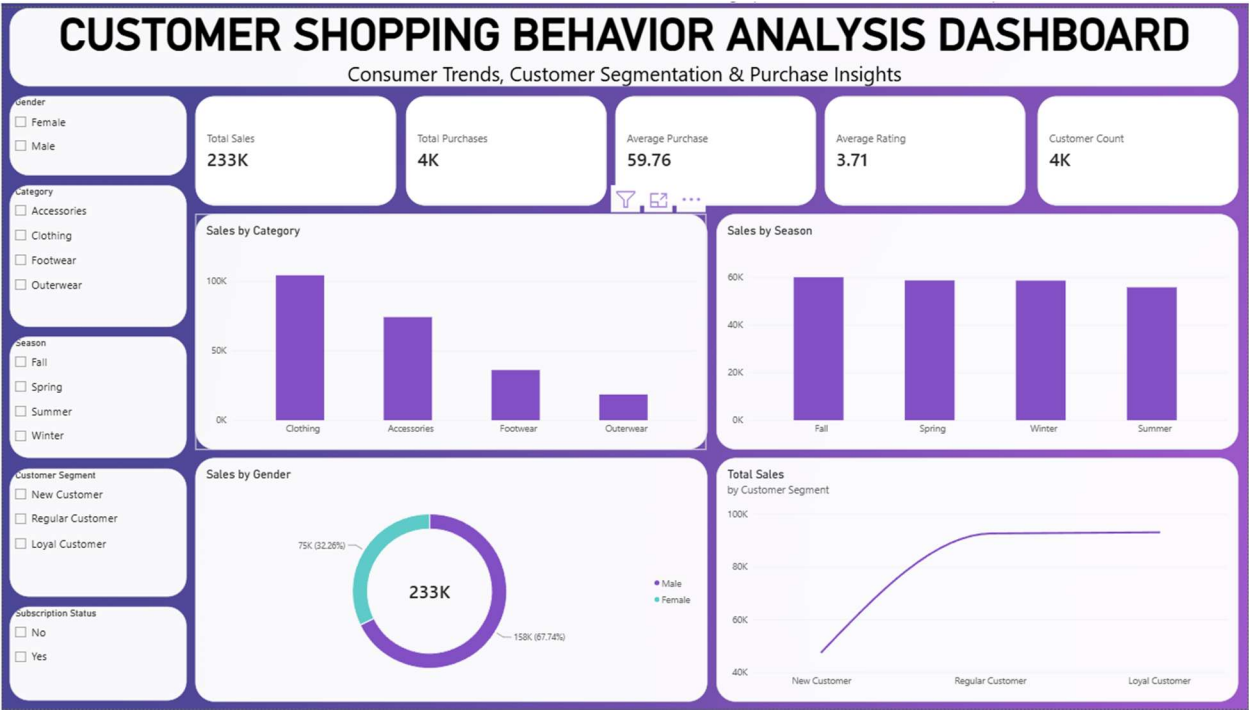


Figure 1: Executive Overview Dashboard

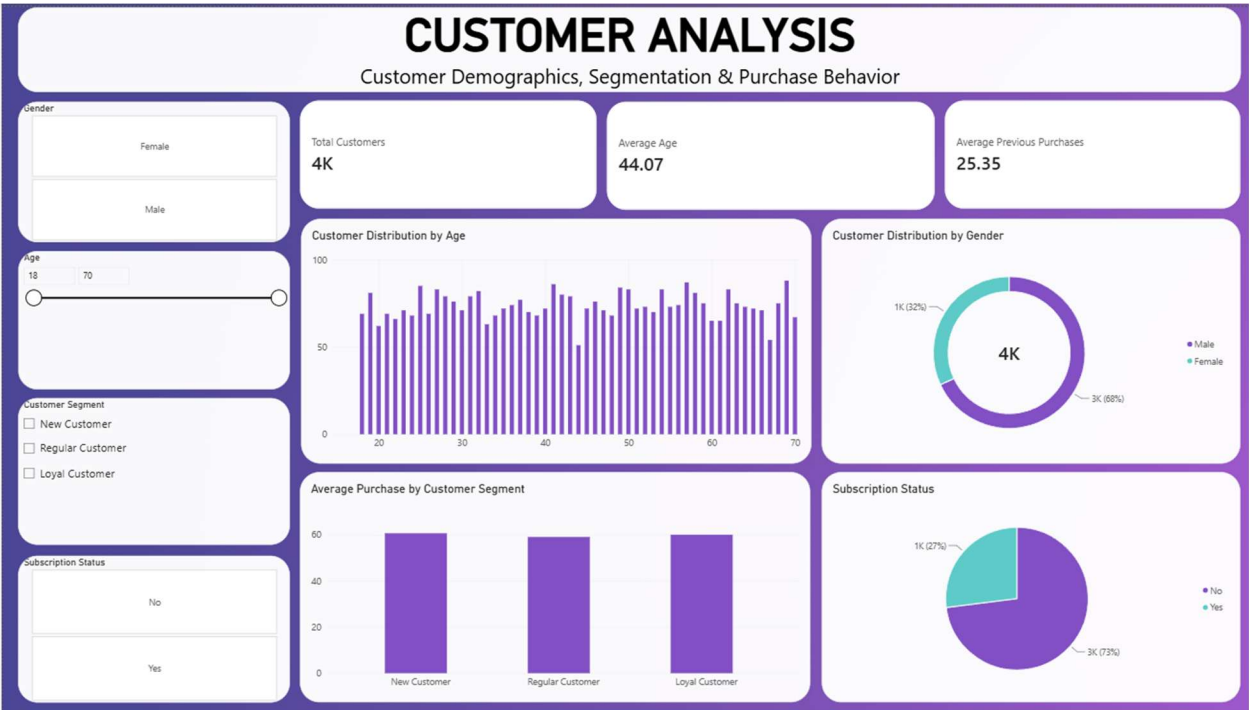


Figure 2: Customer Analysis Dashboard

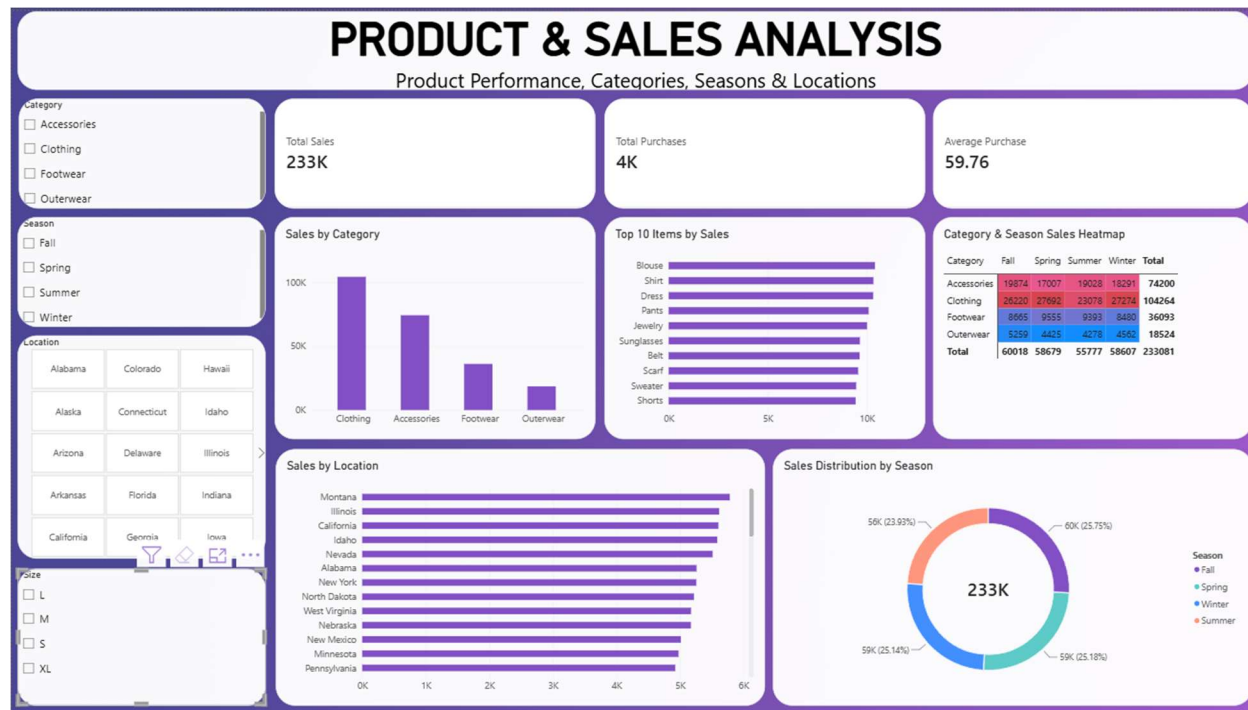


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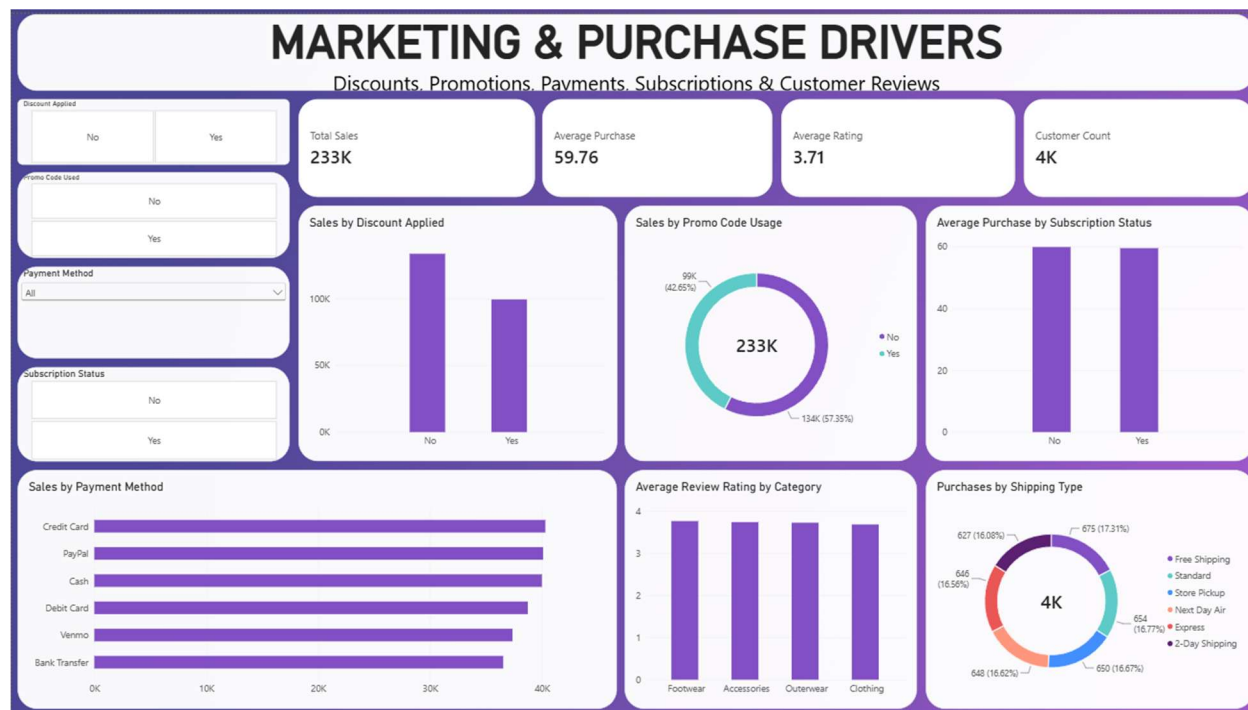


Figure 4: Marketing & Purchase Drivers Dashboard